

Supplementary Text S1

Mathematical Appendix for the Gamma-Code Consensus Model

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This supplementary text contains the mathematical well-posedness results, observable bounds, finite-horizon stability estimates, sufficient selective-preservation re-entry condition, and conservativity discussion referenced in the main manuscript.

Mathematical Appendix

This appendix supplements the GCC with mathematical results that address three common lines of attack: (1) Is the dynamics well-posed? (2) Are the observables controlled? (3) Can the gain-based re-entry idea be formulated as a sufficient condition?

A.1 Assumptions

We consider, over a finite horizon $t \in [0, T]$, the GCC dynamics defined in the main text under the following assumptions.

A1. Structure. $W_{ij}(t) \geq 0$, $\sum_j W_{ij}(t) = 1$ for all i with non-empty rows, $W_{ii}(t) = 0$.

A2. Boundedness. There exist finite constants $G_{\max}$, $\Sigma_{\max}$, $\Phi_{\max}$ such that

$$0 \leq G(t) \leq G_{\max}, \quad 0 \leq g_i(t) \leq 1, \quad 0 \leq \sigma_i(t) \leq \Sigma_{\max}, \quad |\varphi_{ij}(t)| \leq \Phi_{\max}.$$

A3. Temporal regularity. All control functions are measurable and bounded; for stability statements, piecewise continuity suffices.

A4. Observation windows. D_{eff}^λ and M_τ are computed on fixed window lengths $\tau_D > 0$ and $\tau_M > 0$, with the ridge regularization defined in the main text.

A.2 Elementary Bounds of the Observables

Proposition A.1 (Elementary bounds). *For all times t :*

$$0 \leq R(t) \leq 1, \quad 1 \leq D_{\text{eff}}(t) \leq N, \quad 0 \leq M_\tau(t) \leq \frac{1}{4}. \quad (1)$$

Proof sketch. $R(t)$ is the modulus of a mean of unit complex numbers. $D_{\text{eff}}(t)$ is the participation ratio of a positive semi-definite matrix. $M_\tau(t)$ is the variance of a time series with values in $[0, 1]$; its maximum is $\frac{1}{4}$. $\square$

Significance. All three core observables are mathematically compact and controlled. This prevents empirical fits from drifting into unphysical regions.

A.3 Well-Posedness of the Dynamics

Theorem A.2 (Global Lipschitz property and well-posedness). *Define the drift vector $b(\theta, t)$ component-wise by*

$$b_i(\theta, t) = \omega_i + K G(t) g_i(t) \sum_j W_{ij}(t) \sin(\theta_j - \theta_i - \varphi_{ij}(t)). \quad (2)$$

Then, for all $\theta, \psi \in \mathbb{R}^N$ and all $t \in [0, T]$,

$$\|b(\theta, t) - b(\psi, t)\|_\infty \leq 2 |K| G_{\max} \|\theta - \psi\|_\infty. \quad (3)$$

Moreover, $\|b(\theta, t)\|_\infty \leq \|\omega\|_\infty + |K| G_{\max}$. In particular, under A1–A3, the GCC dynamics has a unique global strong solution on $[0, T]$ for every initial condition $\theta(0)$.

Proof sketch. For each component, using $|\sin(x) - \sin(y)| \leq |x - y|$:

$$\begin{aligned} |b_i(\theta, t) - b_i(\psi, t)| &\leq |K| G_{\max} \sum_j W_{ij}(t) |(\theta_j - \theta_i) - (\psi_j - \psi_i)| \\ &\leq |K| G_{\max} \left[\sum_j W_{ij}(t) |\theta_j - \psi_j| + |\theta_i - \psi_i| \underbrace{\sum_j W_{ij}(t)}_{=1} \right] \\ &\leq 2 |K| G_{\max} \|\theta - \psi\|_\infty. \end{aligned} \quad (4)$$

Standard results for SDEs with globally Lipschitz drift and bounded diffusion yield existence and uniqueness. $\square$

A.4 Finite-Horizon Stability

Theorem A.3 (Stability under drift perturbations). *Consider two systems driven by the same Brownian motion:*

$$d\theta(t) = b(\theta(t), t) dt + \Sigma(t) dB(t), \quad (5)$$

$$d\vartheta(t) = \tilde{b}(\vartheta(t), t) dt + \Sigma(t) dB(t). \quad (6)$$

Suppose both drifts are globally L -Lipschitz and

$$\|b(x, t) - \tilde{b}(x, t)\|_\infty \leq \varepsilon(t) \quad \text{for all } x \in \mathbb{R}^N, t \in [0, T]. \quad (7)$$

Then

$$\boxed{\|\theta(t) - \vartheta(t)\|_\infty \leq e^{Lt} \left(\|\theta(0) - \vartheta(0)\|_\infty + \int_0^t \varepsilon(s) ds \right)} \quad (8)$$

Proof sketch. The diffusion terms cancel in the difference equation. For $\delta(t) = \theta(t) - \vartheta(t)$, we have $\|\delta'(t)\|_\infty \leq L \|\delta(t)\|_\infty + \varepsilon(t)$. Gronwall's inequality yields the claim. $\square$

Significance. Small changes in W , G , g_i , or φ_{ij} produce only controlled changes in the trajectory on finite time horizons.

A.5 Continuity of the Observables

Proposition A.4 (Local Lipschitz property). *For fixed τ_D , τ_M , λ , and $\varepsilon_{\text{floor}}$, the observables $R(t)$, $D_{\text{eff}}^\lambda(t)$, and $M_\tau(t)$ are continuous functions of the window trajectory. On compact sets they are locally Lipschitz. In particular, on any suitable compact tube there exists a constant L_{obs} such that*

$$\|O(\text{traj}_1) - O(\text{traj}_2)\|_\infty \leq L_{\text{obs}} \cdot \|\text{traj}_1 - \text{traj}_2\|_\infty, \quad (9)$$

where $O = (R, D_{\text{eff}}^\lambda, M_\tau)$.

Empirical estimation of L_{obs} . One chooses small initial perturbations δ_0 , simulates θ and θ^δ with the same noise path, and computes

$$\hat{L}_{\text{obs}}(\delta_0) = \frac{\sup_t \|O(\theta^\delta)(t) - O(\theta)(t)\|_\infty}{\sup_t \|\theta^\delta(t) - \theta(t)\|_\infty}. \quad (10)$$

A.6 Residual-Backbone Re-Entry

Theorem A.5 (Sufficient condition for re-entry). *Let $S = S(t_1; g_*)$ be the residual backbone structure defined in the main text. Consider, on $I = [t_1, t_2]$, the system restricted to S in which the*

coupling terms coming from outside S are switched off. Define the external residual mass

$$\alpha_S(t) = \max_{i \in S} \sum_{j \notin S} W_{ij}(t). \quad (11)$$

Assume that:

1. The reduced system lies on I within the access region with safety margin $\rho > 0$:

$$\begin{aligned} R_S(t) &> R_{\min} + \rho, \\ D_{\min} + \rho &< D_{\text{eff},S}^\lambda(t) < D_{\max} - \rho, \\ M_{\min} + \rho &< M_{\tau,S}(t) < M_{\max} - \rho. \end{aligned} \quad (12)$$

2. The initial states coincide at t_1 .
3. A local Lipschitz constant L_{obs} for $O = (R_S, D_{\text{eff},S}^\lambda, M_{\tau,S})$ exists.
4. The following holds:

$$\boxed{\int_{t_1}^{t_2} \alpha_S(s) \, ds < \eta, \quad \eta < \frac{\rho}{L_{\text{obs}} \cdot |K| \cdot G_{\max} \cdot e^{2|K|G_{\max}(t_2-t_1)}}} \quad (13)$$

Then the full system restricted to S remains in the access region throughout I .

Proof sketch. By Theorem A.3 with $L = 2|K|G_{\max}$ and $\varepsilon(t) = |K|G_{\max} \alpha_S(t)$,

$$\sup_{t \in I} \|\theta_{\text{full}}(t) - \theta_{\text{red}}(t)\|_\infty \leq |K| G_{\max} e^{2|K|G_{\max}(t_2-t_1)} \cdot \eta. \quad (14)$$

Using L_{obs} , it follows that $\|O_{\text{full}} - O_{\text{red}}\|_\infty \leq L_{\text{obs}} \cdot |K| G_{\max} e^{2|K|G_{\max}(t_2-t_1)} \cdot \eta$. If η is chosen according to (13), this stays below ρ . $\square$

Significance. This is the mathematically sharpest form of the selective-preservation re-entry extension: a transient re-entry is sufficiently possible whenever a residual subnetwork can carry the dynamics and the rest of the system does not perturb it too strongly on a short interval.

Conservativity. The η -condition is a Gronwall-based worst-case bound. If numerical dynamics show a re-entry even though the analytical bound is not satisfied, this primarily reflects conservativity of the bound, not evidence against the simulated mechanism.

A.7 Corollary: Gain-Mediated Lucidity-Like Re-Entry

Corollary A.6. *If there exist a subset S and an interval I such that $\alpha_S(t)$ becomes small on I , the effective drives $\Lambda_i(t) = K G(t) g_i(t)$ for $i \in S$ do not fall below the reduced access threshold, and the*

reduced dynamics on S remains in the access region with safety margin, then a transient re-entry of the full system onto S is internally sufficiently possible within the model.

The mathematical core of the hypothesis:

Lucidity-like re-entry, if generated by the GCC mechanism, can be understood as finite-horizon persistence of a residual backbone regime under decreasing external perturbation load.

A.8 What This Appendix Delivers and What It Does Not

Delivers. Well-posedness of the GCC as an SDE; controlled bounds for the core observables; finite-horizon stability; a genuine sufficient re-entry condition (Theorem [A.5](#)).

Does not deliver. No mathematical proof that real terminal lucidity exists; no general theorem that every form of dementia shifts the access region; no proof of a universal threshold K_c ; no identifiability result for real EEG/MEG data.